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INFUSION SET AUXILIARY DEVICE AND AN INFUSION SET SYSTEM FOR VASCULAR INTERVENTION

Abstract

An infusion set auxiliary device and an infusion set system for vascular intervention are provided. A fixing seat is convex on a bottom plate and has a top surface, a bottom surface and a side surface which is opposite to the bottom plate. The fixing seat includes a plurality of fixing holes arranged along a first direction, and each fixing hole is used for holding a drip pot of one infusion set. The fixing hole penetrates through the side surface, the top surface and the bottom surface to form a first opening, a second opening and a third opening respectively, the sizes of the first opening, the second opening and the third opening satisfy corresponding relationships with the outside diameter of the infusion tube and the outside diameter of the drip pot. The application can arrange multiple infusion sets neatly to facilitate differentiation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of International Application No. PCT/CN2023/126228, filed Oct. 24, 2023, which claims priority to Chinese Patent Application No. 202222850617.5, filed Oct. 27, 2022, the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of vascular intervention medical equipment, and in particular to an infusion set auxiliary device and an infusion set system for vascular intervention used in nerve intervention, cardiovascular intervention, peripheral intervention and other operations.

BACKGROUND

[0003] In a course of vascular interventional therapy, such as nerve intervention, cardiovascular intervention, and peripheral intervention, it may be necessary to use multiple infusion sets to flush the interventional catheter at the same time to prevent intraductal thrombosis or distal vascular embolization events. For example, in neurovascular interventional embolization of cerebral aneurysms, a femoral artery sheath, a guide catheter, an intermediate catheter, a stent micro catheter, and a spring coil embolization catheter these five catheters all need to be flushed with heparin saline. For most vascular interventions, 3-5 infusion sets are required to flush the catheter. Usually, the infusion set includes a bottle stopper puncture set, an infusion tube, a drip pot, a drip rate controller, etc. When the number of the infusion sets is too much, and the infusion tube usually is suspended, resulting in the operation table is messy, it is not suitable to distinguish different infusion sets, and it is not suitable to observe the dripping situation of the drip pots. It is difficult for the operator or assistant to adjust the drip rate for the infusion set that needs to be adjusted, replace the joint, and close the pipe, which brings obstacles to the risk control of surgical operation.

[0004] In addition, due to the number of infusion sets used for flushing catheter is much, and infusion bottles are hung on the infusion rack in a circular distribution, resulting in some of the drip pots being blocked by other infusion tubes and other drip pots, and it is not easy to observe the liquid dripping rate of all drip pots. The drip rate of each infusion set for flushing is manually controlled and the drip rate is subjectively estimated. If the drip rate controller is slipped, the drip rate may be too fast and the fluid entering the catheter may be too much, thus increasing the risk of pulmonary edema, and if the drip rate is slowed or the drip is stopped, intraductal thrombosis may occur, thus increasing the risk of vascular embolism.

[0005] Present application aims to develop an infusion set auxiliary device for vascular intervention and an infusion set system, which provides an infusion set arrangement system for flushing with advantages such as clear marking, orderly arrangement, intuitive observation, and provides an automatic electronic display, adjustment of drip rate and fault alarm system, which can provide flushing convenience for vascular intervention operation, and improve the safety and effectiveness of interventional surgery.

SUMMARY

[0006] present application provides an infusion set auxiliary device and infusion set system for vascular intervention, which is used to organize infusion sets for executing vascular interventional flushing operation.

[0007] In a first aspect, the present application provides an infusion set auxiliary device for vascular intervention, the infusion set auxiliary device includes: [0008] a bottom plate; [0009] a fixing seat, the fixing seat is arranged on the bottom plate, the fixing seat has a top surface, a bottom surface opposite to the top surface, and a side surface connected between the top surface and the bottom surface, the side surface is arranged opposite to the bottom plate, the fixing seat is provided with a plurality of fixing holes arranged in a first direction sequentially, each fixing hole is at least used to contain a drip pot of a disposable medical infusion set; [0010] therein, each fixing hole penetrates through the side surface and forms a first opening on the side surface, and a size of the first opening along the first direction is greater than or equal to the outside diameter of the infusion tube of the infusion set and less than the outside diameter of the drip pot; [0011] therein, each fixing hole penetrates through the top surface and forms a second opening on the top surface, the second opening is connected with the first opening, and a size of the second opening is greater than or equal to the outside diameter of the drip pot; [0012] therein, the fixing hole penetrates through the bottom surface and forms a third opening on the bottom surface, the third opening is connected with the first opening, and a size of the third opening is greater than or equal to the outside diameter of the infusion tube and less than the outside diameter of the drip pot.

[0013] The infusion set auxiliary device provided in present application, through providing a plurality of fixing holes arranged on the fixed base sequentially, and the size of the first opening of each fixing hole is greater than or equal to the outside diameter of the infusion tube and less than the outside diameter of the drip pot, so as to facilitate the infusion tube connected the bottom of the drip pot enters the fixing hole through the first opening, and the size of the second opening of the fixing hole is greater than or equal to the outside diameter of the drip pot, so that the drip pot can be placed in the fixing hole from an upper side of the fixing hole, and the size of the third opening of the fixing hole is greater than or equal to the outside diameter of the infusion tube of the infusion set and smaller than the outside diameter of the drip pot, thus to hold up the drip pot. According to that, the fixing hole can contain the drip pot of the infusion set, and due to the fixing holes are arranged sequentially, the drip pots of multiple infusion sets can be arranged in order, then, the drip pots and infusion tubes of multiple infusion sets are sorted out neatly, which effectively solves the problem that the operation table is messy, and it is not suitable to distinguish different infusion sets, and it is difficult for the operator or assistant to adjust the drip rate, replace the joint, and close the pipe caused by the number of infusion sets are too much.

[0014] In a possible embodiment, the third opening comprises a support opening and a guide channel, the support opening is concentrically arranged with the second opening, the guide channel is connected between the support opening and the first opening, and a size of the support opening is greater than or equal to the outside diameter of the infusion tube and less than the outside diameter of the drip pot. A size of the guide channel along the first direction is greater than or equal to the outside diameter of the infusion tube and less than the size of the support opening.

[0015] In a possible embodiment, the guide channel is deviated from the geometric center of the support opening, and the guide channel is deviated to a right circumferential tangent of the support opening to avoid the infusion tube being pulled directly from the guide channel.

[0016] In a possible embodiment, the bottom plate is provided with a plurality of light-transmitting holes arranged sequentially along the first direction, each light-transmitting hole is connected with one corresponding fixing hole; [0017] the infusion set auxiliary device further includes a plurality of light-emitting elements arranged on a side of the bottom plate deviating from the fixing seat, and each light-emitting element emits light toward to the corresponding fixing hole through the corresponding light-transmitting hole.

[0018] In a possible embodiment, the infusion set auxiliary device further includes a plurality of

light-transmitting glasses, each light-transmitting glass is arranged in one corresponding light-transmitting hole.

[0019] In a possible embodiment, the light-transmitting glass comprises a colorless light-transmitting glass and a colored light-transmitting glass spliced together, the colored light-transmitting glass is adjacent to the third opening relative to the colorless light-transmitting glass, and the colored light-transmitting glass is used for setting at a position at least relative to liquid at the bottom of the drip pot.

[0020] In a possible embodiment, the colors of the colored light-transmitting glasses in at least two of the light-transmitting holes are different, and the colors of the colored light-transmitting glasses may comprise: red, green, blue, purple, orange.

[0021] In a possible embodiment, the infusion set auxiliary device further includes a drip controller, the drip controller is located outside the infusion tube at the top of the drip pot to electronically control the drip rate of the drip in the drip pot instead of a traditional manual control device for the drip rate of the infusion set.

[0022] In a possible embodiment, the infusion set auxiliary device further includes at least one drip detector each arranged on an inner wall of the corresponding fixing hole, and the drip detector is set relative to the drip of the drip pot to detect the drip rate of the drip in the drip pot.

[0023] In a possible embodiment, the infusion set auxiliary device further includes at least one liquid detector each sets with respect to an infusion tube connected to the top of the drip pot, and the liquid detector is used for detecting whether the infusion tube at the top of the drip pot has liquid; [0024] the infusion set auxiliary device also comprises an alarm, the alarm is electrically connected with the drip detector and the liquid detector, the alarm is used to generate an alarm signal when the drip rate detected by the drip detector is 0 and the liquid in the infusion tube detected by the liquid detector is empty; [0025] the infusion set auxiliary device further comprises a display device that electrically connects the drip detectors and the liquid detectors, and the display device is used to display the drip rate in the drip pot and display information about whether there is liquid in the infusion tube connected to the top of the drip pot.

[0026] In a second aspect, the application further provides an infusion set system for vascular intervention, the infusion set system includes a plurality of infusion sets and the above infusion set auxiliary device, each infusion set includes a drip pot arranged in the fixing hole of the infusion set auxiliary device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] In order to more clearly illustrate the technical scheme of embodiments of the present application, brief introduction of the drawings required to be used in the embodiments are presented below.

[0028] FIG. 1 is a structural diagram of an infusion set auxiliary device for vascular intervention provided in present application;

[0029] FIG. 2 is a structural diagram of an infusion set system for vascular intervention provided in present application;

[0030] FIG. 3 is a top view of the infusion set auxiliary device provided in FIG. 1;

[0031] FIG. 4 is an upward view of the infusion set auxiliary device provided in FIG. 1;

[0032] FIG. 5 is a section view of the infusion set auxiliary device along a X-Z plane provided in FIG. 1;

[0033] FIG. 6 is a rear view of the infusion set auxiliary device provided in FIG. 1;

[0034] FIG. 7 is a circuit structure block diagram of the infusion set auxiliary device provided in FIG. 1.

ILLUSTRATION FOR LABELS OF THE DRAWINGS

[0035] infusion set auxiliary device—**100**; Bottom plate—**10**; Fixing seat—**20**; Top surface—**201**; Bottom surface—**202**; Side surface—**203**; fixing hole—**21**; Infusion set—**200**; Drip pot—**210**; First opening—**211**; Infusion tube—**220**; Second opening—**212**; Third opening—**213**; Support opening—**213a**; Guide channel—**213b**; Light—transmitting hole—**101**; Light source box—**30**; Light—emitting element—**31**; Light—transmitting glass—**40**; Colorless light—transmitting glass—**41**; Colored light—transmitting glass—**42**; Drip detector—**50**; Liquid detector—**60**; Alarm—**70**; Display device—**80**; Processor—**90**.

DETAILED DESCRIPTION

[0036] The technical scheme in the embodiments of present application will be clearly and completely described in combination with the drawings attached to the embodiments of present application. Obviously, the embodiments described are only part of the embodiments of present application, but not all embodiments. In addition, the phrase “embodiment” in present application means that a particular feature, structure or characteristic described in conjunction with an embodiment may be included in at least one embodiment of present application. The presence of the phrase “embodiment” at various positions in the specification does not refer to the same embodiment, and is also not independent embodiment or alternative embodiment that is mutually exclusive with other embodiments. It is understood, both explicitly and implicitly, by those skilled in the art that, the embodiments described in the present application may be combined with other embodiments.

[0037] In some vascular interventional therapy scenarios, multiple infusion sets are used simultaneously to flush the interventional catheter, for example, in neurovascular interventional embolization of cerebral aneurysms, a femoral artery sheath, a guide catheter, an intermediate catheter, a stent micro catheter, and a spring coil embolization catheter these five catheters all need to be flushed with heparin saline. For most vascular interventions, 3-5 infusion sets are required to flush the catheter. Usually, the infusion set includes a bottle stopper puncture set, an infusion tube, a drip pot, a drip rate controller, etc. When the number of the infusion sets is too much, and the infusion tube usually is suspended, resulting in the operation table is mussy, it is not suitable to distinguish different infusion sets, and it is not suitable to observe the dripping situation of the drip pots. It is difficult for the operator or assistant to adjust the drip rate for the infusion set that needs to be adjusted, replace the joint, and close the pipe, which brings obstacles to the risk control of surgical operation. Present application aims to develop an infusion set auxiliary device for vascular intervention and an infusion set system, which provides an infusion set arrangement system for flushing with advantages such as clear marking, orderly arrangement, intuitive observation, and provides an automatic electronic adjustment of drip rate and fault alarm system, which can provide flushing convenience for vascular intervention operation, and improve the safety and effectiveness of interventional surgery.

[0038] Please refer to FIG. 1, present application provides an infusion set auxiliary device **100** for vascular intervention, the infusion set auxiliary device **100** includes a bottom plate **10** and a fixing seat **20**.

[0039] The bottom plate **10** may be plate-like, and its shape is not specified. Alternately, the shape of the bottom plate **10** is rectangular. The volume of the bottom plate **10** is not specified in present application. Alternately, the bottom plate **10** is placed upright as shown in FIG. 1, the volume of the bottom plate **10** is relatively small, e.g. 195 mm (millimeter) width*110 mm height*4 mm thickness. The infusion set auxiliary device **100** also includes a cross bar with a length of 200 mm. A cross bar connection point is arranged on a rear surface (the surface away from the fixing seat **20**) of the bottom plate **10** to facilitate the subsequent installation of a light box (the light box is described in the following section), and also facilitate to fix an interventional surgical rod in an neurovascular interventional therapy.

[0040] The fixing seat **20** is convexly arranged on the bottom plate **10**. The bottom plate **10** and the

fixing seat **20** can be formed integrately. The shape of the fixing seat **20** is not specified.

Alternately, the fixed seat **20** is rectangular.

[0041] In order to facilitate description, in FIG. **1**, a X-axis direction is defined as the width direction (forward or backward direction) of the fixing seat **20**, a Y-axis direction is defined as the length direction (left-right direction) of fixing seat **20**, and a Z-axis direction is defined as the height direction (up-down direction) of the fixing seat **20**.

[0042] The fixing seat **20** has a top surface **201**, a bottom surface **202** opposite to the top surface **201**, and a side surface **203** connected between the top surface **201** and the bottom surface **202**. The side surface **203** is arranged opposite to the bottom plate **10**. The top surface **201** faces the top side/upper side, that is, the positive direction of the Z axis, and the bottom surface **202** faces the bottom side, that is, the negative direction of the Z axis. The side surface **203** faces the front side, that is, the positive direction of the X-axis.

[0043] Please referring to FIG. **1** and FIG. **2**, the fixing seat **20** is provided with a plurality of fixing holes **21** arranged along a first direction sequentially. The first direction is the Y-axis direction. Each fixing hole **21** is at least used for containing a drip pot **210** of a disposable medical infusion set **200**. The present application does not make a specific limit on the number of fixing holes **21**, for example, the number of fixing holes **21** may be 2, 3, 4, 5, 6 or more than 6, etc.

[0044] Each fixing hole **21** is set along the Z-axis direction, namely is extended along the Z-axis direction, so that the drip pot **210** is arranged along the vertical direction. Alternately, the fixing hole **21** can roughly be a cylindrical hole.

[0045] Referring to FIG. **1** and FIG. **2**, the fixing hole **21** penetrates through the side surface **203** and forms a first opening **211** on the side surface **203**, and a size (width) of the first opening **211** along the first direction is greater than or equal to the outside diameter of an infusion pipe **220** and is smaller than the outside diameter of the drip pot **210**, so that the infusion pipe **220** connected to the bottom of the drip pot **210** is facilitated to enter the fixing hole **21** through the first opening **211**.

[0046] Referring to FIG. **1** and FIG. **3**, the fixing hole **21** penetrates through the top surface **201** and forms a second opening **212** on the top surface **201**, the second opening **212** is connected to the first opening **211**, and a size of the second opening **212** is greater than or equal to the outside diameter of the drip pot **210**, so that the drip pot **210** can be placed in the fixing hole **21** from the upper side of the fixing hole **21**.

[0047] Referring to FIG. **1** and FIG. **4**, the fixing hole **21** penetrates through the bottom surface **202** and forms a third opening **213** on the bottom surface **202**, the third opening **213** is connected with the first opening **211**, a size of the third opening **213** is greater than or equal to the outside diameter of the infusion pipe **220** of the infusion apparatus **200** and is smaller than the outside diameter of the drip pot **210**, so that the infusion pipe **220** of the infusion apparatus **200** can pass through the third opening **213**, and the bottom wall of the fixing hole **21** can hold up the drip pot **210**, and the drip pot **210** thus is contained in the fixing hole **21**.

[0048] When placing the infusion set **200** on the infusion set auxiliary device **100**, first, the infusion tube **220** at the bottom of the drip pot **210** enters the fixing hole **21** through the first opening **211**, and then the drip pot **210** enters the fixing hole **21** from the upper side of the fixing hole **21** until the bottom of the drip pot **210** is connected to the bottom wall of the fixing hole **21**, so that the drip pot **210** is arranged in the fixing hole **21** of the fixing seat **20**. The drip pots **210** of multiple infusion sets **200** are arranged in different fixing holes **21** in the above way, so as to arrange the infusion tubes **220** and drip pots **210** of different infusion set **200**, thus facilitate to quickly adjust the drip rate, replace the joint, and close the pipe for the infusion set **200** that is needed. It also avoids some of drip pots **210** are blocked by other infusion tubes **220** and other drip pots **210**, thus facilitate to observe the liquid dripping rate of all drip pots.

[0049] The fixing seat **20** provided in this embodiment is provided with 5 fixing holes **21b** arranged neatly, which can organize 1-5 infusion sets **200** (one to five infusion sets **200**) and facilitate the arrangement and use of the 1-5 infusion sets **200**.

[0050] The first opening **211** can be used as an observation window to observe the liquid in the drip pot **210** and the drip rate of the drip pot **210**. In present application, the observation window is a long rectangle set along the Z axis. The present application does not specify the width of the observation window along the Y-axis direction, such as, the width of the observation window can be 10 mm. The width of the first opening **211** is less than the outer diameter of the drip pot **210** to avoid the drip pot **210** contained in the fixing hole **21** being removed from the first opening **211**.

[0051] The size of the second opening **212** are not specified in present application. Alternately, the size of the second opening **212** may be larger than the outside diameter of drip pots **210** of all sizes, thus, any drip pot **210** can be inserted into the fixing hole **21** from above, and the infusion set auxiliary device **100** provided in present application can be used with all types of infusion sets **200**.

[0052] Alternately, the size of the second opening **212** is 25 mm (but not limited to this size), which can be compatible with the infusion sets **200** commonly used from different manufacturers, thus the second opening **212** has strong compatibility.

[0053] The infusion set auxiliary device **100** provided in present application, through providing a plurality of fixing holes **21** arranged on the fixed base **20** sequentially, and the size of the first opening **211** of each fixing hole **21** is greater than or equal to the outside diameter of the infusion tube **220** and less than the outside diameter of the drip pot **210**, so as to facilitate the infusion tube **220** connected the bottom of the drip pot **210** enters the fixing hole **21** through the first opening **211**, and the size of the second opening **212** of the fixing hole **21** is greater than or equal to the outside diameter of the drip pot **210**, so that the drip pot **210** can be placed in the fixing hole **21** from an upper side of the fixing hole **21**, and the size of the third opening **213** of the fixing hole is greater than or equal to the outside diameter of the infusion tube **220** of the infusion set **200** and smaller than the outside diameter of the drip pot **210**, thus to hold up the drip pot **210**. According to that, the fixing hole **21** can contain the drip pot **210** of the infusion set **200**, and due to the fixing holes **21** are arranged sequentially, the drip pots **210** of multiple infusion sets **200** can be arranged in order, then, the drip pots **210** and infusion tubes **220** of multiple infusion sets **200** are sorted out neatly, which effectively solves the problem that the operation table is messy, and it is not suitable to distinguish different infusion sets **200**, and it is difficult for the operator or assistant to adjust the drip rate, replace the joint, and close the pipe, caused by the number of infusion sets **200** are too much.

[0054] Alternately, please referring to FIG. 4, the third opening **213** includes a support opening **213a** and a guide channel **213b**. The support opening **213a** is concentrically arranged with the second opening **212**. Optionally, the support opening **213a** and the second opening **212** may both be circular. A center of the support opening **213a** is collinear with a center of the second opening **212** and the collinear is arranged/extended along the Z-axis direction, namely a connection line between the center of the support opening **213a** and the center of the second opening **212** is arranged/extended along the Z-axis direction.

[0055] A size of the support opening **213a** is greater than or equal to the outside diameter of the infusion tube **220** and less than the outside diameter of the drip pot **210**. Alternately, the bottom of the drip pot **210** may be arranged in the support opening **213a** to fix the drip pot **210** in the fixing hole **21**. When the inner diameter of the fixing hole **21** is greater than the outside diameter of the drip pot **210**, the drip pot **210** can be contained and fixed in the fixing hole **21** by placing the bottom of the drip pot **210** in the support opening **213a**, so as to prevent the tilt or instability of the drip pot **210** in the fixing hole **21**.

[0056] Alternately, an inner wall of the support opening **213a** may be conical to fit the outer contour of the bottom of the drip pot **210**, thus to improve the stability of the drip pot **210** in the fixing hole **21**.

[0057] Alternately, the support opening **213a** may be a cylindrical hole with a diameter of 10 mm (but not limited to that size). The support opening **213a** can support the bottom of drip pots **210** of all sizes, so that to fix the drip pots **210** of any size.

[0058] The guide channel **213b** is connected between the support opening **213a** and the first opening **211**. The size of the guide channel **213b** along the first direction is greater than or equal to the outside diameter of the infusion tube **220** and less than the dimensions of the support opening **213a**. The infusion tube **220** enters the fixed hole **21** through the first opening **211** and then enters the support opening **213a** through the guide channel **213b**.

[0059] For example, the guiding channel **213b** has a width direction size of 3 mm (but not limited to that size) to facilitate the infusion tube **220** to pass through.

[0060] The present application does not specify the extension direction of guide channel **213b**. Alternately, the guide channel **213b** can be set/extended along the X axis.

[0061] Alternately, the guide channel **213b** is deviated from the geometric center of the support opening **213a**. The center line of the guide channel **213b** is not collinear with the geometric center of the support opening **213a**, so that the infusion tube **220** contained in the support opening **213a** is not easy to move out through the guide channel **213b**, and the containment stability of the fixing hole **21** is improved.

[0062] Please referring to FIG. 4, the guide channel **213b** is deviated to a right circumferential tangent of the support opening **213a** to avoid the infusion tube **220** being pulled directly from the guide channel **213b**.

[0063] In other embodiments, the guide channel **213b** may also be extended in a inclined direction, or in a direction of a bend line, or in a direction of a curve, etc., so that the infusion tube **220** located in the supporting opening **213a** is not easy to move out through the guide channel **213b**, thus improving the containment stability of the fixing hole **21**.

[0064] Alternately, the size of the first opening **211** along the first direction is greater than the size of the guide channel **213b** along the first direction.

[0065] Generally, the material of infusion set **200** is mostly made of colorless and transparent chemical materials, and the commonly used interventional catheter irrigation solution is colorless and transparent normal saline. As a result, the drip and liquid level in drip pot **210** cannot be clearly identified, which may lead to excessive dripping speed, excessive cardiac load, and lead to the air enters into blood vessels and resulting in the risk of air embolism.

[0066] The infusion set auxiliary device **100** provided in present application can also clearly distinguish the drip and liquid levels in the drip pot **210**.

[0067] For details, please referring to FIG. 1 and FIG. 5, the bottom plate **10** is provided with a plurality of light-transmitting holes **101** arranged sequentially along the first direction, and each light-transmitting hole **101** is connected with a fixing hole **21**. Therein, the light-transmitting hole **101** is in the shape of a long strip and is arranged along the Z-axis direction.

[0068] Please referring to FIG. 5, the infusion set auxiliary device **100** further includes a light box **30**. The light box **30** is arranged on one side of the bottom plate **10** away from the fixing seat **20**. A plurality of light-emitting elements **31** are provided in the light box **30**. The light-emitting elements **31** are arranged on a side, of the bottom plate **10**, deviating from the fixing seat **20**, and plurality of light-emitting elements **31** are corresponded to the plurality of fixing hole **21** one by one, and are corresponded to plurality of light-transmitting holes **101** one by one, and each light-emitting element **31** emits light toward the corresponding fixing hole **21** through the corresponding light-transmitting hole **101**.

[0069] Alternately, the light-emitting element **31** include but are not limited to LED light beads, LED light strips, etc. Optionally, the light emitted by the light-emitting element **31** is white light.

[0070] The present application does not specify the size of light-transmitting holes **101**. Alternately, the width of light-transmitting holes **101** along the Y axis is 3.5 mm, and the height of the light-transmitting holes **101** along the Z axis is 50 mm.

[0071] Please referring to FIG. 5 and FIG. 6, the infusion set auxiliary device **100** further includes a plurality of light-transmitting glass **40**. Each light-transmitting glass **40** is arranged in a light-transmitting hole **101**. The light-transmitting glass **40** is embedded in the light-transmitting hole

101.

[0072] Please referring to FIG. 5 and FIG. 6, the light-transmitting glass **40** comprises colorless light-transmitting glass **41** and colored light-transmitting glass **42** spliced together. The colored light-transmitting glass **42** is adjacent to the third opening **213** relative to the colorless light-transmitting glass **41**. This application does not specify the connection boundary of colorless light-transmitting glass **41** and colored light-transmitting glass **42**. Alternately, the area/size of colored translucent glass **42** is greater than that of colorless translucent glass **41**. For example, the colorless light-transmitting glass **41** with a height of 15 to 20 mm is installed on the upper part of light transmission hole **101**, and the colored light-transmitting glass **42** with a height of 30 to 35 mm is installed on the bottom part of light-transmission hole **101**.

[0073] The colored light-transmitting glass **42** is set at a position relative to liquid at the bottom (lower) part of the drip pot **210**. The colorless light-transmitting glass **41** is set at a position relative to the top (upper) part of the drip pot **210**.

[0074] The colored light-transmitting glass **42** is a glass that has a certain color and allows the light to pass through, namely transparent or semitransparent. A part of the light emitted by the light-emitting element **31** is transmitted to the liquid of the drip pot **210** through the colored light-transmitting glass **42**. The height of the liquid in the drip pot **210** can be clearly observed by using the refraction phenomenon of the liquid, and at this time, the liquid shows the color of the colored light-transmitting glass **42**. The other part of the light emitted by the luminescent element **31** is directly transmitted to the drip of the drip pot **210** through the colorless light-transmitting glass **41**, so that the drip in the drip pot **210** can be clearly observed.

[0075] Further, the colors of the colored light-transmitting glasses **42** in at least two of the light-transmitting holes **101** are different, so that the liquid in at least two drip pot **210** may appear different colors, the colors of the colored light-transmitting glasses **42** may comprise: red, green, blue, purple, orange.

[0076] Alternately, the colors of colored light-transmitting glass **42** in different light transmission holes **101** are different, that is, each colored light-transmitting glass **42** in the corresponding light transmission hole **101** has a unique color, so that the liquid in different drip pot **210** can be of a different color to facilitate to distinguish different infusion sets **200**.

[0077] Alternately, the colors of the colored light-transmitting glass **42** in the two adjacent light-transmitting holes **101** are different, so that the liquid in the two adjacent drip pots **210** can appear different colors, so as to facilitate to distinguish two adjacent infusion sets **200**.

[0078] In addition, number labels can be set at corresponding positions of the fixing holes **21** sequentially to distinguish different drip pots **210** by different number labels, and facilitate the adjustment of the operators or facilitate the operators to communicate with each other and assist the adjustment.

[0079] The infusion set auxiliary device **100** provided in present application, can easily and intuitively distinguish drip and liquid levels in the drip pot **210**, so as to timely respond to abnormal drip and liquid levels in the drip pot **210** and reduce therapy risks. In addition, it can also make the drip pots **210** of different infusion sets **200** are highly distinguishable, which can be neatly arranged from left to right, and can identify different drip pots **210** by using different number labels, and can also distinguish different drip pot **210** by using different colors, which is convenient for the operators to adjust or communicate with each other and assist the adjustment.

[0080] The present application does not specify the material of the bottom plate **10** and the fixing seat **20**. Alternately, the material of the bottom plate **10** and the fixing seat **20** include but is not limited to plastic, metal, stainless steel, etc. In this embodiment, the bottom plate **10** and the fixing seat **20** are made of stainless steel matte material, which are prepared by laser cutting technology and are well-made, thus is conducive to long-term use.

[0081] The drip rate of a general infusion set **200** usually is manually controlled, and the amount of infusion per minute is estimated according to experience, which is not standardized, resulting in

inaccurate drip rate; In addition, when the amount of suspension fluid, namely the fluid in the infusion bottle, reaches 500 ml, it is sometimes necessary to pressurize, and when the pressure cannot be stopped in time after the liquid is transferred, the risk of air embolism may occur.

[0082] In one possible embodiment, the infusion set auxiliary device **100** further includes a drip controller (not shown), which is located outside the infusion tube **220** at the top of the drip pot **210** to electronically control the drip rate of the drip in the drip pot **210** instead of a traditional manual control device for the drip rate of the infusion set.

[0083] Referring to FIG. 7, the infusion set auxiliary device **100** for vascular intervention provided in present application also includes at least one drip detector **50**. Each drip detector **50** is arranged on the inner wall of the corresponding fixing hole **21**. The drip detector **50** is set relative to the drip of the drip pot **210** to detect the drip rate of the drip in the drip pot **210**. Alternately, the number of the at least one drip detector **50** is equal to the number of the fixing hole **21**, and each drip detector **50** corresponds with one fixing hole **21**.

[0084] Alternately, the drip detector **50** includes, but is not limited to, photoelectric non-contact sensors, such as infrared sensors, which include a transmitter and a receiver, respectively located on opposite sides of the inner wall of the fixed hole **21**, and the transmitter and receiver are respectively located on opposite sides of drip pot **210** and near the top of the drip pot **210**. The detection principle of infrared sensor includes but is not limited to: the light emitted by the transmitter passes through a drop path of the drip, and the light emitted by the transmitter is received by the receiver when the light emitted by the emitter does not pass through the drip. When the light emitted by the emitter passes through the drip, the receiver does not receive the light from the emitter, and the infrared sensor detects the dynamics of the drip, the drip rate per minute, and the drip rate is more accurate and standardized than the results determined by humans.

[0085] The infrared sensor can be powered by silicon photovoltaic cells or other batteries.

[0086] Please referring to FIG. 7, the infusion set auxiliary device **100** further includes at least one liquid detector **60**. Each liquid detector **60** can be arranged on the inner wall of the fixing hole **21** or outside the fixing hole **21**. When the infusion tube **220** at the top of the drip pot **210** is contained in fixed hole **21**, the liquid detector **60** may be arranged on the inner wall of fixing hole **21** and arranged on opposite sides of the infusion tube **220** at the top of the drip pot **210**, or the liquid detector **60** may be clamped on opposite sides of the infusion tube **220** at the top of the drip pot **210**. When the infusion tube **220** at the top of the drip pot **210** is positioned outside the fixing hole **21**, the liquid detector **60** is clamped on opposite sides of the infusion tube **220** at the top of the drip pot **210**. Alternately, the number of the at least one liquid detector **60** is equal to the number of the fixing hole **21**, and each liquid detector **60** corresponds with one fixing hole **21**.

[0087] The liquid detector **60** is arranged relative to the infusion tube **220** connected to the top of the drip pot **210**. The liquid detector **60** is used to detect whether the infusion tube **220** at the top of the drip pot **210** has a liquid. The liquid detector **60** includes, but is not limited to, photoelectric non-contact sensors, such as infrared sensors, whose detection principle may be the same or different from that of the drip detector **50**.

[0088] Alternately, please referring to FIG. 7, the infusion set auxiliary device **100** further includes an alarm **70**. The alarm **70** is electrically connected to the drip detector **50** and the liquid detector **60**. The alarm **70** is used to generate an alarm signal when the drip rate detected by the drip detector **50** is 0 and the liquid in the infusion tube **220** detected by the liquid detector **60** is empty. Alternately, the alarm **70** is electrically connected to all of the drip detector **50** and all of the liquid detector **60**, and the alarm **70** generates the alarm signal when the drip rate detected by the drip detector **50** corresponding to one fixing hole **21** is 0 and the liquid in the infusion tube **220** detected by the liquid detector **60** corresponding to the same fixing hole **21** is empty.

[0089] Alternately, the alarm **70** includes, but is not limited to, an audible alarm or a warning light. When the drip rate detected by the drip detector **50** is 0 and the liquid in the infusion tube **220** detected by the liquid detector **60** is empty, the alarm **70** generates a light or audible warning so

that the operator can observe that the drip rate detected by the drip detector **50** is 0 and the liquid in the infusion tube **220** detected by the liquid detector **60** is empty. Therefore, corresponding operations can be executed by the operator in time to avoid air embolism.

[0090] Alternately, please referring to FIG. 7, the infusion set auxiliary device **100** further includes a display device **80**. The display device **80** is electrically connected to the drip detector **50** and the liquid detector **60**. The display device **80** is used to display the drip rate in the drip pot **210** and information about whether there is liquid in the infusion tube **220** connected to the top of the drip pot.

[0091] The infusion set auxiliary device **100** further comprises a processor **90**, the display device **80**, the drip detector **50** and the liquid detector **60** are all electrically connected to the processor **90**. The drip detector **50** and the liquid detector **60** are used for sending corresponding detection result to the processor **90**, the processor **90** processes the test results and controls the display device **80** to display the drip rate in the drip pot **210** and information about whether there is liquid in the infusion tube **220** connected to the top of the drip pot, after processing the test results, so that the user or the operator can observe such information directly. Alternately, the processor **90** is electrically connected to all of the drip detector **50** and all of the liquid detector **60**, and the processor **90** display the drip rate in the drip pot **210** contained in one fixing opening **21** and the information about whether there is liquid in the infusion tube **220** connected to the top of the drip pot **210** contained in the same fixing opening **21** together.

[0092] The number of the display device **80** can be one or more, for example, the number of the display device **80** can be two, one is located in the ward, the other is located in the work station of the medical staff, so that the medical staff can monitor the infusion situation in real time.

[0093] The infusion set auxiliary device **100** provided in present application may include either or both of the alarm **70** and the display device **80**.

[0094] Please referring back to FIG. 2. the present application also provides an infusion set system **1000** for vascular intervention, the infusion set system **1000** includes a plurality of infusion sets **200** and the infusion set auxiliary equipment **100** described in any of the above embodiments. The infusion set **200** comprises a drip pot **210**, which is arranged in the fixing hole **21** of the infusion set auxiliary device **100**, and the infusion set auxiliary device **100** can arrange a plurality of drip pot **210** of the infusion set **200** neatly, and facilitate to organize multiple infusion sets **200**, make it easy to distinguish the drip and liquid level in the drip pots **210** of multiple infusion sets **200**, and accurately obtain the situation that the drip rate of the drip pot **210** is 0 and the liquid of the infusion tube **220** at the top of the drip pot **210** is empty, and can send an alarm signal to remind medical staff to respond in time and reduce the risk of air embolism.

[0095] The above descriptions are only optional implementations of the disclosure, and are not intended to limit the present disclosure. Any modifications, equivalent replacements, improvements, etc. made within the spirit and principles of the present disclosure shall be included in the protection range of the present disclosure.

Claims

1. An infusion set auxiliary device for vascular intervention, wherein the infusion set auxiliary device comprises: a bottom plate; a fixing seat, arranged on the bottom plate, the fixing seat has a top surface, a bottom surface opposite to the top surface, and a side surface connected between the top surface and the bottom surface, the side surface is arranged opposite to the bottom plate, the fixing seat is provided with a plurality of fixing holes arranged in a first direction sequentially, each fixing hole is at least used to contain a drip pot of an infusion set; wherein each fixing hole penetrates through the side surface and forms a first opening on the side surface, and a size of the first opening along the first direction is greater than or equal to the outside diameter of an infusion tube of the infusion set and less than the outside diameter of the drip pot; wherein each fixing hole

penetrates through the top surface and forms a second opening on the top surface, the second opening is connected with the first opening, and a size of the second opening is greater than or equal to the outside diameter of the drip pot; wherein the fixing hole penetrates through the bottom surface and forms a third opening on the bottom surface, the third opening is connected with the first opening, and a size of the third opening is greater than or equal to the outside diameter of the infusion tube and less than the outside diameter of the drip pot.

2. The infusion set auxiliary device of claim 1, wherein the third opening comprises a support opening and a guide channel, the support opening is concentrically arranged with the second opening, the guide channel is connected between the support opening and the first opening, and a size of the support opening is greater than or equal to the outside diameter of the infusion tube and less than the outside diameter of the drip pot, a size of the guide channel along the first direction is greater than or equal to the outside diameter of the infusion tube and less than the size of the support opening.

3. The infusion set auxiliary device of claim 2, wherein the guide channel is deviated from the geometric center of the support opening.

4. The infusion set auxiliary device of claim 2, wherein the guide channel is extended in a inclined direction, or in a direction of a bend line, or in a direction of a curve.

5. The infusion set auxiliary device of claim 2, wherein the size of the first opening along the first direction is greater than the size of the guide channel along the first direction.

6. The infusion set auxiliary device of claim 1, wherein the bottom plate is provided with a plurality of light-transmitting holes arranged sequentially along the first direction, each light-transmitting hole is connected with one corresponding fixing hole; wherein the infusion set auxiliary device further comprises a plurality of light-emitting elements arranged on a side of the bottom plate deviating from the fixing seat, and each light-emitting element emits light toward to the corresponding fixing hole through the corresponding light-transmitting hole.

7. The infusion set auxiliary device of claim 6, wherein the infusion set auxiliary device further comprises a plurality of light-transmitting glasses, each light-transmitting glass is arranged in one corresponding light-transmitting hole.

8. The infusion set auxiliary device of claim 7, wherein the light-transmitting glass comprises a colorless light-transmitting glass and a colored light-transmitting glass spliced together, the colored light-transmitting glass is adjacent to the third opening relative to the colorless light-transmitting glass, and the colored light-transmitting glass sets at a position at least relative to liquid at a bottom of the drip pot.

9. The infusion set auxiliary device of claim 8, wherein colors of the colored light-transmitting glasses in at least two of the light-transmitting holes are different.

10. The infusion set auxiliary device of claim 8, wherein colors of the colored light-transmitting glasses in different light transmission holes are different.

11. The infusion set auxiliary device of claim 8, wherein colors of the colored light-transmitting glasses in two adjacent light-transmitting holes are different.

12. The infusion set auxiliary device of claim 1, wherein the infusion set auxiliary device further comprises at least one drip detector each arranges on an inner wall of the corresponding fixing hole, and the drip detector is set relative to the drip of the corresponding drip pot to detect the drip rate of the drip in the drip pot.

13. The infusion set auxiliary device of claim 12, wherein the infusion set auxiliary device further comprises at least one liquid detector each sets with respect to an infusion tube connected to a top of the corresponding drip pot, and the liquid detector is used for detecting whether the infusion tube at the top of the drip pot has liquid.

14. The infusion set auxiliary device of claim 13, wherein the infusion set auxiliary device further comprises an alarm, the alarm is electrically connected with the drip detector and the liquid detector and is used to generate an alarm signal when the drip rate detected by the drip detector is 0

and the liquid in the infusion tube detected by the liquid detector is empty.

15. The infusion set auxiliary device of claim 13, wherein the infusion set auxiliary device further comprises a display device, the display device is electrically connected to the drip detector and the liquid detector, and the display device is used to display the drip rate in the drip pot and display information about whether there is liquid in the infusion tube connected to the top of the drip pot.

16. An infusion set system for vascular intervention, wherein the infusion set system comprises a plurality of infusion sets and an infusion set auxiliary device, each infusion set comprises a drip pot; wherein the infusion set auxiliary device comprises: a bottom plate; a fixing seat, arranged on the bottom plate, the fixing seat has a top surface, a bottom surface opposite to the top surface, and a side surface connected between the top surface and the bottom surface, the side surface is arranged opposite to the bottom plate, the fixing seat is provided with a plurality of fixing holes arranged in a first direction sequentially, each fixing hole is at least used to contain the drip pot of one infusion set; wherein each fixing hole penetrates through the side surface and forms a first opening on the side surface, and a size of the first opening along the first direction is greater than or equal to the outside diameter of an infusion tube of the infusion set and less than the outside diameter of the drip pot; wherein each fixing hole penetrates through the top surface and forms a second opening on the top surface, the second opening is connected with the first opening, and a size of the second opening is greater than or equal to the outside diameter of the drip pot; wherein the fixing hole penetrates through the bottom surface and forms a third opening on the bottom surface, the third opening is connected with the first opening, and a size of the third opening is greater than or equal to the outside diameter of the infusion tube and less than the outside diameter of the drip pot.

17. The infusion set system of claim 16, wherein the third opening comprises a support opening and a guide channel, the support opening is concentrically arranged with the second opening, the guide channel is connected between the support opening and the first opening, and a size of the support opening is greater than or equal to the outside diameter of the infusion tube and less than the outside diameter of the drip pot, a size of the guide channel along the first direction is greater than or equal to the outside diameter of the infusion tube and less than the size of the support opening.

18. The infusion set system of claim 17, wherein the guide channel is deviated from the geometric center of the support opening.

19. The infusion set system of claim 17, wherein the guide channel is extended in a inclined direction, or in a direction of a bend line, or in a direction of a curve.

20. The infusion set system of claim 17, wherein the size of the first opening along the first direction is greater than the size of the guide channel along the first direction.
